

John Kim

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EDUCATION

University of Virginia

B.S. in Computer Science, School of Engineering and Applied Sciences

- GPA: 4.0/4.0
- Activities: ML@UVA Operations Officer

Charlottesville, VA

Aug. 2025 – May 2028

Thomas Jefferson High School for Science and Technology

Advanced Studies Diploma

- GPA: 4.463/4.0; AP 5s in CS A, Calculus BC, Physics C (Mechanics & E&M), and Chemistry

Alexandria, VA

Aug. 2021 – Jun. 2025

EXPERIENCE

Software Engineer

May 2026 – Present

N2O (Deaver Varney LLC), Enterprise & Government Software Consultancy

Remote

- Developed a learning content platform to replace an enterprise client's commercial system (client under NDA)
- Owned client projects end-to-end (design, implementation, testing, and deployment), rotating to a new system every few weeks, working directly with the CTO on a small distributed engineering team

Researcher, ML@UVA & Johns Hopkins APL

Sep. 2025 – Present

University of Virginia

Charlottesville, VA

- Researched agentic AI and human-aligned decision-making in LLM agents through the strategy game *Diplomacy*
- Designed benchmarks to evaluate LLM agent performance in multi-agent gameplay
- Applied steering vectors and prompting techniques to align agent behavior with distinct human playstyles

Full Stack Developer Intern

Jun. 2025 – Aug. 2025

MySmaX (AIoT Startup), Seoul National University

Seoul, South Korea

- Designed and deployed an AI agent that automates workflows for any IoT device connected to MySmaX's JOI platform, using Model Context Protocol (MCP)
- Applied machine learning-based anomaly detection to analyze real-time IoT device telemetry
- Built MySmaX's production website frontend in Next.js as primary contributor, implementing Figma designs

Competitive Programming & Mathematics

2021 – Present

Clubs & Competitions

Charlottesville, VA

- USACO Gold Division
- American Invitational Mathematics Examination (AIME) Qualifier (4x)
- Jane Street Estimathon Winner (2024, 2025)
- Active member of the Putnam Club and ICPC Club at UVA

PROJECTS

Hide in Plain Sight! (Roblox Game) (Play) | *Lua, Roblox Studio, Blender*

Jun. 2026 – Present

- Solo-developed and shipped a cross-platform (PC, mobile, tablet) multiplayer hide-and-seek game with 8 worlds
- Engineered a wall- and ceiling-climbing character controller and a surface-camouflage paint system in Lua; modeled the 3D characters in Blender
- Reached the top 10,000 Roblox games within a month of release, with 68K+ visits and 150 peak concurrent players

Alpine Ski Racing AI Analysis Model (GitHub) | *Python, YOLO, PyTorch, CNNs*

Aug. 2024 – May 2025

- Built a CNN-based computer vision pipeline that extracts skiers' pose data from race videos to give technique feedback
- Validated the model against real races, showing technique scores correlated with finish times; documented in a TJHSST senior research paper

Offline AI Model for North Korea (Website) | *Python, LangChain, HuggingFace, Unsloth*

Jun. 2024 – May 2025

- Fine-tuned LLMs to build an offline generative AI system for securely distributing information in North Korea
- Received a \$7,500 grant from the Human Rights Foundation recognizing innovation and potential social impact

TECHNICAL SKILLS

Languages: Python, Java, C++, JavaScript, TypeScript, Lua, HTML, CSS

Frameworks: React, Next.js, Django, Tailwind CSS, MCP, FastAPI

Developer Tools: Git, Docker, Linux Shell, Blender, Jira, AWS

Libraries: PyTorch, TensorFlow, pandas, LangChain, HuggingFace, Agno, Unsloth

Spoken Languages: English (Native), Korean (Native), Latin (Advanced) | **Other:** Alpine ski racer & junior ski coach